

Online Participatory Budgeting

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Outline

- 1 Introduction to Online Optimization
- 2 Introduction to Participatory Budgeting
- 3 Online Participatory Budgeting

Online LP

Borodin and El-Yaniv [2005]

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Object x come 1 by 1



Online Knapsack/ Secretary problem

Marchetti-Spaccamela and Vercellis [1995], Bateni et al. [2013]

- Utility function f
- Constraint (Cardinal or Knapsack)
- Look for approx guarantees

Results

- no constant approx Marchetti-Spaccamela and Vercellis [1995], Chakrabarty et al. [2008]
- $1/10e$ approx by Babaioff et al. [2007] for random order
- $1/2e$ approx by Vaze [2017] for small individual contributions

The setting

Budget $b = 50$

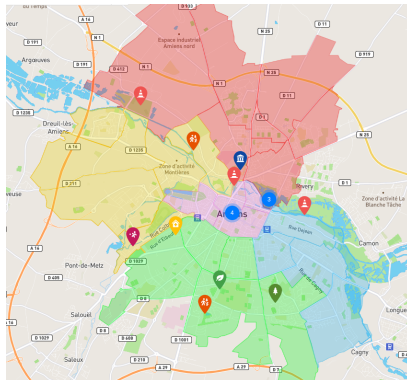
project	cost	votes
chocolate cake	50	
carrot cake	20	
banana bread	30	

Most common type of vote $\rightarrow$ approval ($A = A_1, \dots, A_n$)

How to take votes into account ?

Welfarist rules
(score maximization)

- Approval Voting: number of approved projects
- Proportional Approval Voting: $1 + \frac{1}{2} + \dots + \frac{1}{AV}$



Method of Equal Shares

Peters et al. [2021]

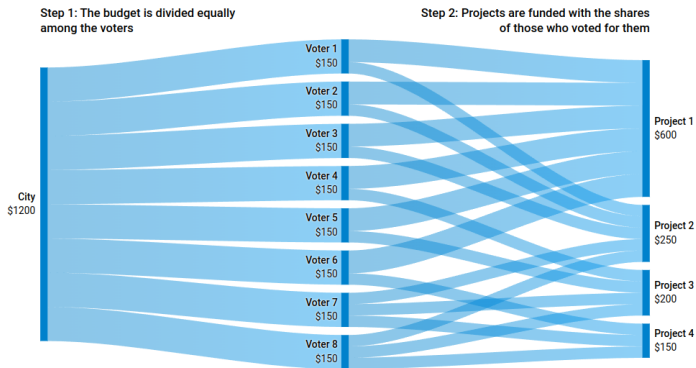


Figure: <https://equalshares.net>

How to evaluate an outcome ?

Rey and Maly [2023]

Definition

Priceability: Possible to explain outcome by a contribution graph

Definition

Local Fair Share: equal contribution function $\Rightarrow$ no more project affordable

Definition

Cohesiveness: For $S \subseteq N$, $Q \subseteq P$ if $\frac{|S|}{n} \geq \frac{c(Q)}{b}$ and $\forall i \in S, Q \subseteq A_i$
Proportionality: S is Q -cohesive, $\text{sat}_S(\pi) \geq \text{sat}_S(Q)$.

About proportionality

Definition

Satisfaction Function: mapping from $\mathcal{P}(P)$ to $\mathbb{R}$

Different variations

- Perfect Justified Representation: $\text{sat}_S(\pi) = \min_{i \in S} \text{sat}(A_i \cap \pi)$
Not always possible
- Extended Justified Representation: $\text{sat}_S(\pi) = \max_{i \in S} \text{sat}(A_i \cap \pi)$
- Proportional Justified Representation:
 $\text{sat}_S(\pi) = \text{sat}(\bigcup_{i \in S} A_i \cap \pi)$

Results for Offline PB

Axiom	Priceability	PJR-1	EJR-1	LFS
MES	✓	✓	✓	✗
MES(share)	✓	✓	✓	✓
PAV	✗	✗	✗	✗

The framework

From Online ABC elections Do et al. [2022]

Example: vote for funding missions

project	cost	voter 1	voter 2	voter 3
IJCAI (South Korea)	3k	✓	✓	✓
ECAI (Spain)	1k	✗	✓	✓
SAGT (Amsterdam)	1k	✓	✗	✓

NB: We can have predictions

The Greedy Budgeting rule

Split budget and pay for affordable projects

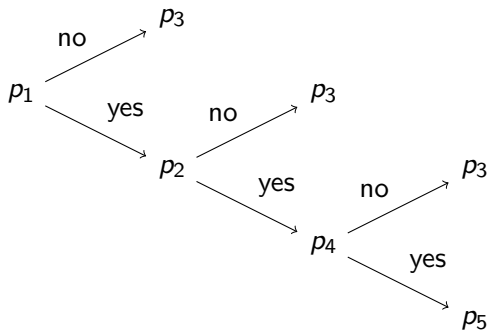
- Satisfies Priceability (then $\text{PJR}[\text{cost}]-1$)
- 2 versions
- Introducing Efficient Priceability

project	1	2	3	price
p_1	1	0	1	$5k$
p_2	1	0	1	$2k$
p_3	0	1	1	$4k$
p_4	1	0	0	$1k$

Priceability and Local FS

budget $b = 6001$

project	1	2	3	price
p_1	0	0	1	$2k$
p_2	0	1	1	$2k$
p_3	0	0	0	$3k$
p_4	1	1	0	$2k$
p_5	1	0	0	$1k$



Approximating axioms

Definition

α -Cohesiveness: For $S \subseteq N$, $Q \subseteq P$ if $\frac{|S|}{n} \geq \alpha \frac{c(Q)}{b}$ and

$\forall i \in S, Q \subseteq A_i$

Proportionality: S is α -Q-cohesive, it deserves $\text{sat}_S(\pi) \geq \text{sat}_S(Q)$ of the budget.

	JR	PJR	EJR
sat^{card}	JR $[\text{sat}^{\text{card}}]$	γ_b - PJR $[\text{sat}^{\text{card}}]$	γ_b - EJR $[\text{sat}^{\text{card}}]$
sat^{cost}	2-JR $[\text{sat}^{\text{cost}}]$	2-PJR $[\text{sat}^{\text{cost}}]$	τ_b - EJR $[\text{sat}^{\text{cost}}]$

$$b/\tau_b(\log_2(b/\tau_b) + 1) = b = \lfloor b/\gamma_b H(b/\gamma_b) \rfloor$$

$$b = 1000 \implies \gamma_b \simeq \tau_b = 8$$

Conclusion

Pros

Some optimal fairness guarantees

Online better than repeated one-shot on PJR[cost]

Cons/

optimal $\nrightarrow$ good

Further

How to integrate predictions ?

Questions

Perpetual PB

JR+ properties

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